Chapter 12: Physical Storage Systems

Outline

- Classification of Physical Storage Media
- Storage Hierarchy
- Magnetic Disks
  - Disk Interface Standards
  - Performance Measures of Disks
  - Optimization of Disk-Block Access
- Flash Storage & SSD
- Storage Class Memory(NVM)

Classification of Physical Storage Media

Can differentiate storage into:

- **volatile storage(易失存储):** loses contents when power is switched off
- **non-volatile storage(非易失存储)**:
  - Contents persist even when power is switched off.
  - Includes secondary and tertiary storage, as well as batter-backed up main-memory.

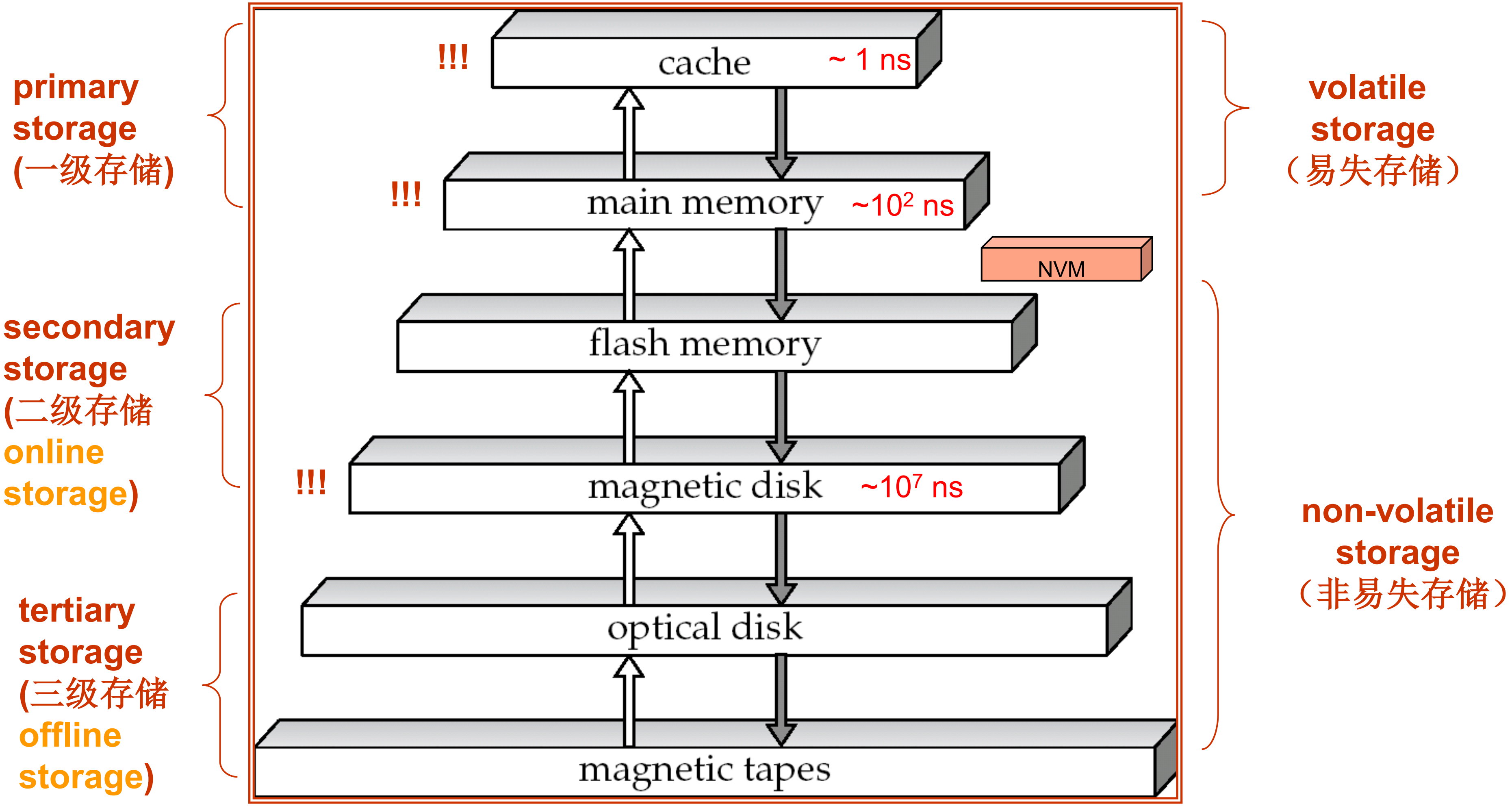
Speed with which data can be accessed

Cost per unit of data

Reliability

- data loss on power failure or system crash
- physical failure of the storage device

Storage Hierarchy（存储级别）



Storage Hierarchy (Cont.)

primary storage: Fastest media but volatile (cache, main memory).

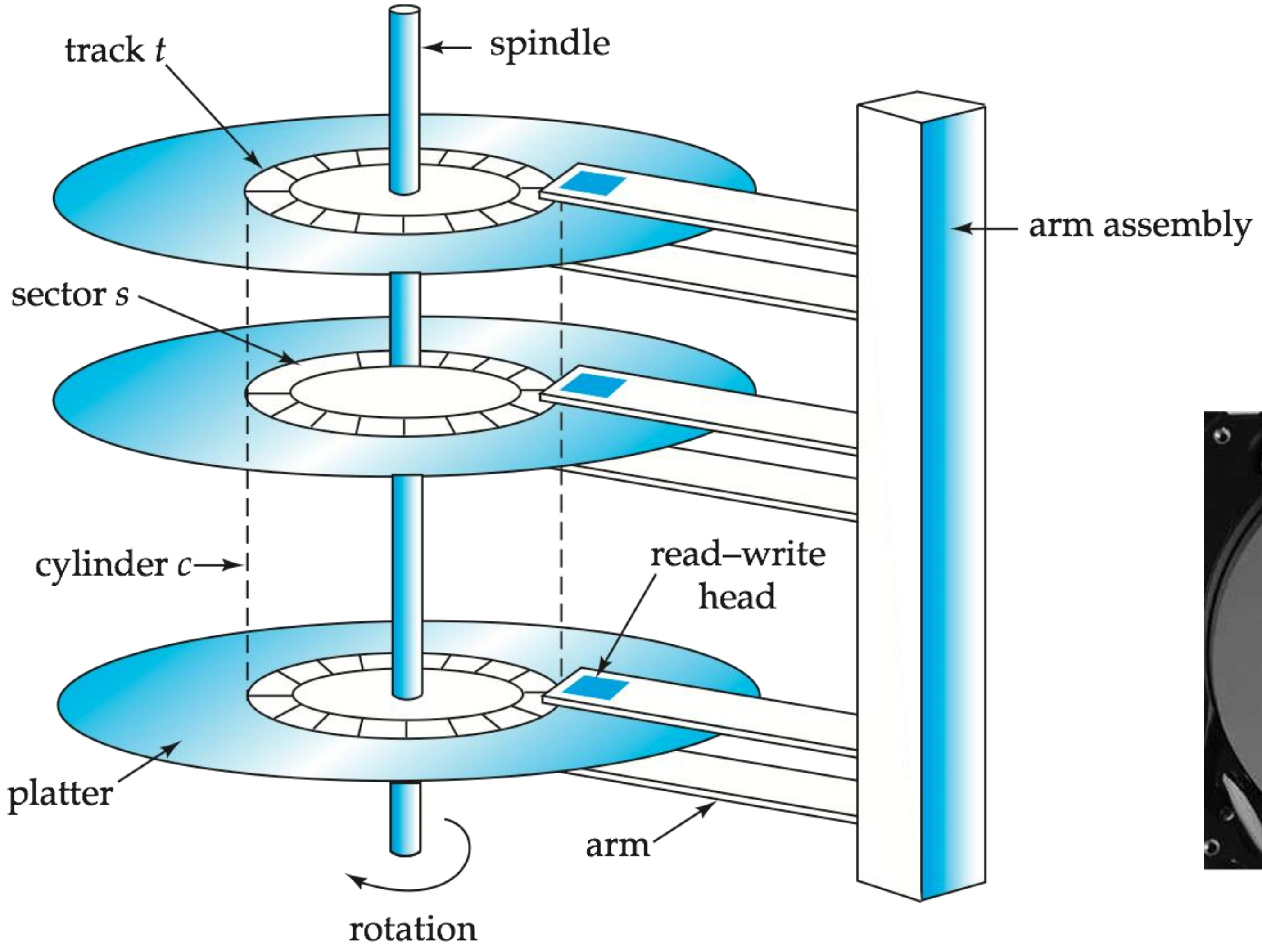
secondary storage: next level in hierarchy, non-volatile, moderately fast access time

- also called **on-line storage**
- E.g. flash memory, magnetic disks

tertiary storage: lowest level in hierarchy, non-volatile, slow access time

- also called **off-line storage**
- E.g. magnetic tape, optical storage

Magnetic Hard Disk Mechanism



Simplified Diagram of Disk Drives

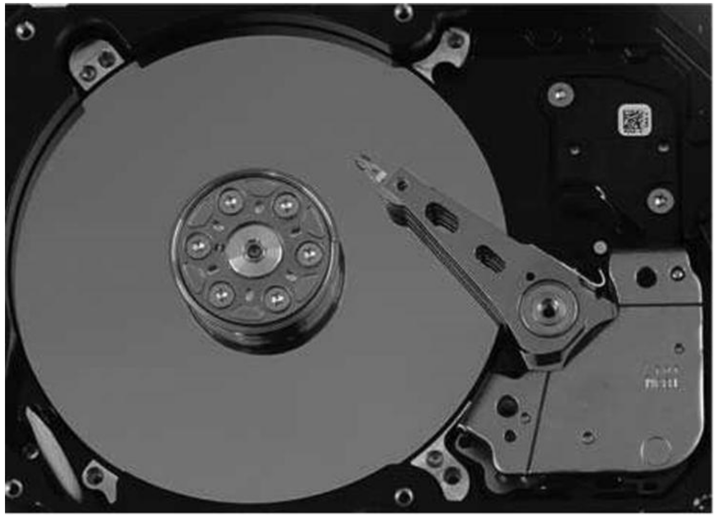


Photo of magnetic disk drive

Magnetic Disks

Read-write head

- Positioned very close to the platter surface (almost touching it)
- Reads or writes magnetically encoded information.

Surface of platter divided into circular **tracks（磁道）**

- Over 50K-100K tracks per platter on typical hard disks

Each track is divided into **sectors（扇区）.**

- A sector is the smallest unit of data that can be read or written.
- Sector size typically 512 bytes
- Typical sectors per track: 500 to 1000 (on inner tracks) to 1000 to 2000 (on outer tracks)

To read/write a sector

- disk arm swings to position head on right track
- platter spins continually; data is read/written as sector passes under head

Head-disk assemblies

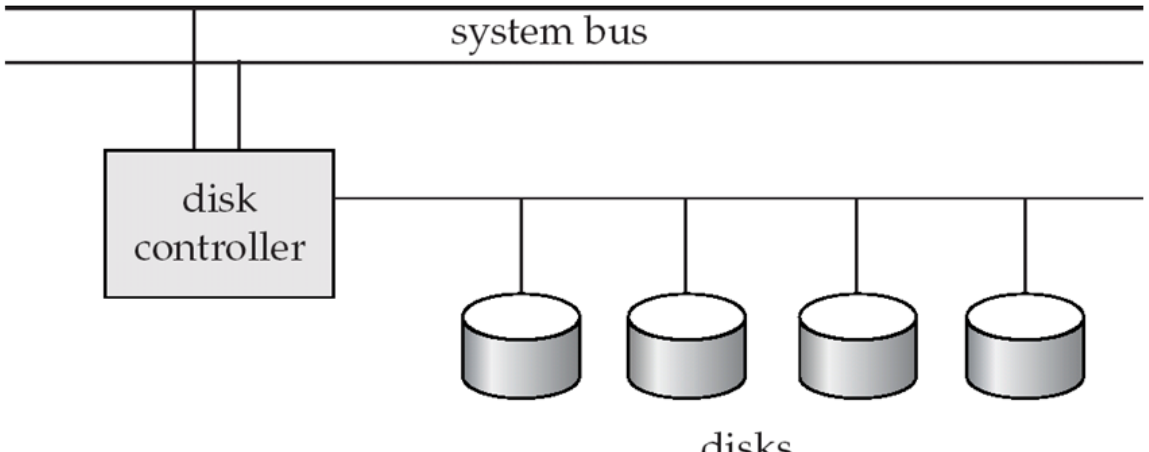
- multiple disk platters on a single spindle (1 to 5 usually)
- one head per platter, mounted on a common arm.

Cylinder（柱面） *i* consists of *i*th track of all the platters

Magnetic Disks (Cont.)

Disk controller(磁盘控制器) – interfaces between the computer system and the disk drive hardware.

- accepts high-level commands to read or write a sector
- initiates actions such as moving the disk arm to the right track and actually reading or writing the data
- Computes and attaches **checksums** to each sector to verify that data is read back correctly
  - If data is corrupted, with very high probability stored checksum won't match recomputed checksum
- Ensures successful writing by reading back sector after writing it
- Performs **remapping of bad sectors**



Performance Measures of Disks

Access time(访问时间) – the time it takes from when a read or write request is issued to when data transfer begins. Consists of:

- **Seek time（寻道时间）** – time it takes to reposition the arm over the correct track.
  - Average seek time is 1/2 the worst case seek time.
  - 4 to 10 milliseconds on typical disks
- **Rotational latency（旋转延迟)**– time it takes for the sector to be accessed to appear under the head.
  - Average latency is 1/2 of the worst case latency.
  - 4 to 11 milliseconds on typical disks (5400 to 15000 r.p.m.)

Data-transfer rate（数据传输率） – the rate at which data can be retrieved from or stored to the disk.

- 25 to 100 MB per second max rate, lower for inner tracks
- Multiple disks may share a controller, so rate that controller can handle is also important
  - E.g. SATA: 150 MB/sec, SATA-II 3Gb (300 MB/sec)
  - Ultra 320 SCSI: 320 MB/s, SAS (3 to 6 Gb/sec)
  - Fiber Channel (FC2Gb or 4Gb): 256 to 512 MB/s

Performance Measures of Disks（Cont.）

Disk block is a logical unit for storage allocation and retrieval

- 4 to 16 kilobytes typically
  - Smaller blocks: more transfers from disk
  - Larger blocks: more space wasted due to partially filled blocks

Sequential access pattern(顺序访问模式)

- Successive requests are for successive disk blocks
- Disk seek required only for first block

Random access pattern（随机访问模式）

- Successive requests are for blocks that can be anywhere on disk
- Each access requires a seek
- Transfer rates are low since a lot of time is wasted in seeks

I/O operations per second (IOPS，每秒I/O操作数)

- Number of random block reads that a disk can support per second
- 50 to 200 IOPS on current generation magnetic disks

Performance Measures (Cont.)

Mean time to failure (MTTF，平均故障时间) – the average time the disk is expected to run continuously without any failure.

- Typically 3 to 5 years
- Probability of failure of new disks is quite low, corresponding to a “theoretical MTTF” of 500,000 to 1,200,000 hours（57 to 136 years）for a new disk
- An MTTF of 1,200,000 hours for a new disk means that given 1000 relatively new disks, on an average one will fail every 1200 hours(50 days)
- MTTF decreases as disk ages

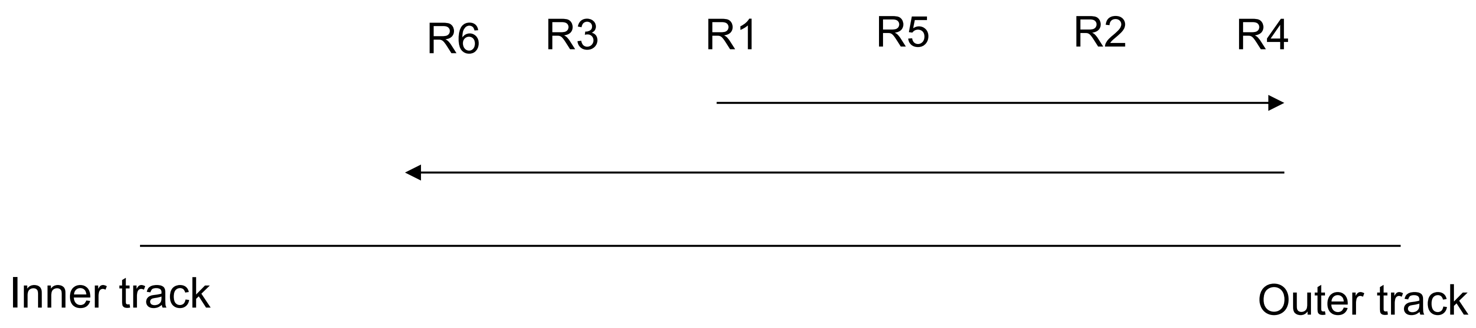
Optimization of Disk-Block Access

Buffering: in-memory buffer to cache disk blocks

Read-ahead(Prefetch): Read extra blocks from a track in anticipation that they will be requested soon

Disk-arm-scheduling algorithms re-order block requests so that disk arm movement is minimized

elevator algorithm



Optimization of Disk Block Access (Cont.)

File organization

Allocate blocks of a file in as contiguous a manner as possible

Allocation in units of **extents(盘区）**

Files may get **fragmented**

- E.g., if free blocks on disk are scattered, and newly created file has its blocks scattered over the disk
- Sequential access to a fragmented file results in increased disk arm movement
- Some systems have utilities to **defragment** the file system, in order to speed up file access

Optimization of Disk Block Access (Cont.)

Nonvolatile write buffers（非易失性写缓存） – speed up disk writes by writing blocks to a non-volatile RAM buffer immediately

- Non-volatile RAM: battery backed up RAM or flash memory
  - Even if power fails, the data is safe and will be written to disk when power returns
- Controller then writes to disk whenever the disk has no other requests or request has been pending for some time
- Database operations that require data to be safely stored before continuing can continue without waiting for data to be written to disk
- *Writes can be reordered to minimize disk arm movement*

Log disk（日志磁盘） – a disk devoted to writing a sequential log of block updates

- Used exactly like nonvolatile RAM
  - Write to log disk is very fast since no seeks are required
  - No need for special hardware (NV-RAM)

Flash Storage

NAND flash - used widely for storage, cheaper than NOR flash

- requires page-at-a-time read (page: 512 bytes to 4 KB)
  - Not much difference between sequential and random read
- Page can only be written once
  - Must be erased to allow rewrite

SSD(Solid State Disks)

- Use standard block-oriented disk interfaces, but store data on multiple flash storage devices internally

	Magnetic Disk	**Solid State Disk**
Retrieve a page	5-10 milliseconds	20-100 microseconds
Random access	Random 50 to 200 IOPS	reads: 10,000 IOPS writes: 40,000 IOPS
Data transfer rate	200M	500M(SATA), 3G(NVMe)
Power consumption	higher	lower
Update mode	in place	erase → rewrite
Realiability	MTTF : 500,000 to 1,200,000 hours	erase blocks: 100,000 to 1,000,000 erases

Flash Storage (Cont.)

Erase happens in units of **erase block**

  Takes 2 to 5 milliseconds

  Erase block typically 256 KB to 1 MB (128 to 256 pages)

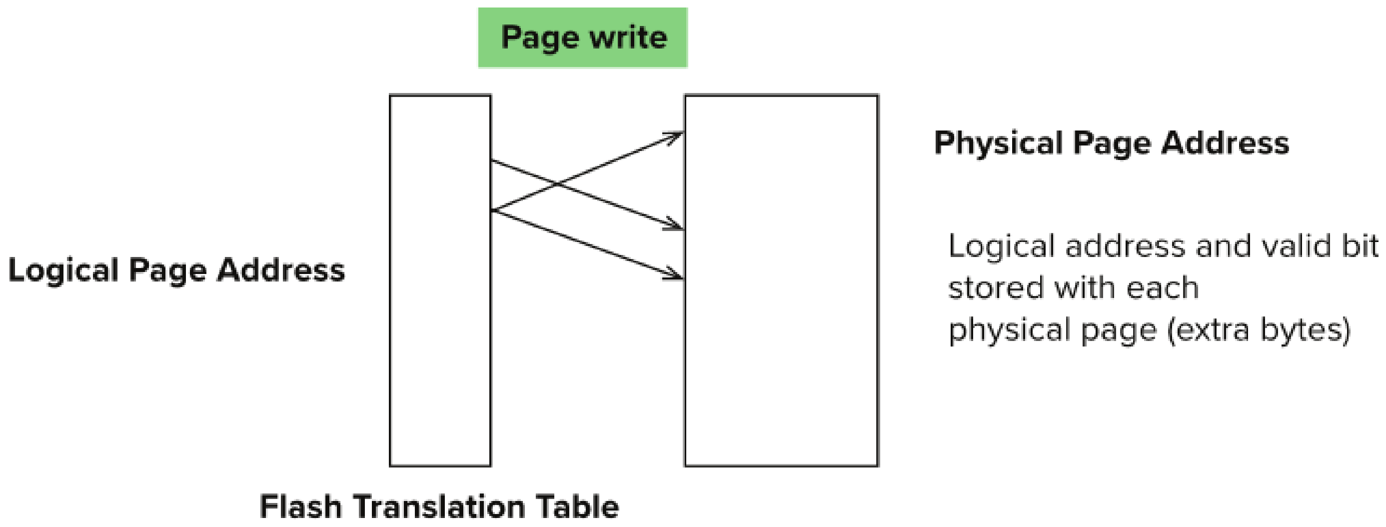
  After 100,000 to 1,000,000 erases, erase block becomes unreliable and cannot be used

Remapping of logical page addresses to physical page addresses avoids waiting for erase

Flash translation table tracks mapping

  also stored in a label field of flash page

  remapping carried out by **flash translation layer**



wear leveling(磨损均衡)- evenly distributed erase operators across physical blocks

Storage Class Memory(NVM)

3D-XPoint memory technology pioneered by Intel

Available as Intel Optane

- **SSD interface** shipped from 2017
  - Allows lower latency than flash SSDs
- **Non-volatile memory interface** announced in 2018
  - Supports direct access to words, at speeds comparable to main-memory speeds

	DRAM	NVM	SSD	HDD
Read Latency	1x	2 - 4 x	500x	10^5 x
Write Latency	1x	2 - 8 x	5000x	10^5 x
Persistence	No	Yes	Yes	Yes
Byte-Addressable	Yes	Yes	No	No
Endurance	Yes	No	No	Yes

End of Chapter 12